SIMATS ENGINEERING

ASSESSMENT TOOL 3: Class test

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: ***Apply image enhancement and segmentation techniques for extracting meaningful regions from images.*** **(BTL 3)**

QUESTIONS: 10 **Max Marks50** **Duration :60 Minutes**

Annexure A

Class test

Code of Conduct:

I A. Midhuna (192521302) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

A. Midhuna

Signature of the student:

Annexure A

Class test

1. A nighttime traffic surveillance camera captures images with insufficient brightness, making vehicles and pedestrians difficult to recognize. Apply suitable gamma correction or logarithmic transformation techniques to improve image visibility and explain the enhancement procedure.

2. An underwater imaging system captures coral reef images with uneven illumination and limited contrast. Apply histogram equalization or Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance the visibility of underwater structures.

3. A digital camera captures an image containing salt-and-pepper noise that affects the identification of important objects. Apply a suitable spatial filtering technique to remove the noise while preserving object boundaries and explain the processing steps.

4. A printed circuit board inspection system captures images in which small component boundaries appear blurred. Apply Laplacian sharpening or unsharp masking to improve the visibility of component edges and explain the sharpening procedure.

5. An astronomical imaging system captures distant celestial objects with unwanted high-frequency noise. Apply an appropriate frequency-domain low-pass filtering technique to suppress noise and improve the visibility of important astronomical features.

6. A textile inspection system captures fabric images containing subtle thread patterns that are difficult to distinguish because of slowly varying background illumination. Apply a suitable frequency-domain high-pass filtering technique to enhance the fine texture details and identify irregularities.

7. An automated metal surface inspection system must identify isolated defective pixels and narrow scratches on manufactured components. Apply suitable point detection and line detection techniques to locate these defects and explain the detection procedure.

8. A document processing system captures handwritten examination answer sheets under uneven lighting conditions. Apply an appropriate adaptive thresholding technique to separate handwritten characters from the background and improve text extraction.

9. A satellite-based environmental monitoring system must identify flood-affected regions with similar intensity characteristics. Apply a suitable region-growing segmentation technique to group connected pixels and extract the affected areas.

10. An automated blueprint analysis system processes engineering drawings containing disconnected lines and incomplete object boundaries. Apply appropriate edge linking and boundary detection techniques to reconstruct continuous structural outlines and explain the processing steps.

Rubrics for Evaluation

Criteria	*Weightage*	*Level 4 – Exemplary*	*Level 3 – Proficient*	*Level 2 – Developing*	*Level 1 – Beginning*
Understanding of Concepts	*25%*	*Demonstrates thorough, accurate understanding of concepts*	*Good understanding with minor gaps*	*Basic understanding with some misconceptions*	*Poor understanding; major misconceptions*
Explanation	*25%*	*Detailed, well-explained answers with relevant examples*	*Clear explanation with adequate details*	*Limited explanation; lacks depth*	*Poor or unclear explanation*
Application	*20%*	*Effectively applies concepts with strong analytical ability*	*Applies concepts with minor errors*	*Limited application with weak analysis*	*No application or incorrect analysis*
Organization	*15%*	*Well-structured answers with logical flow and coherence*	*Organized with minor issues in flow*	*Basic structure, lacks clarity*	*Poor organization, incoherent answers*
Accuracy	*10%*	*Completely accurate and covers all required points*	*Mostly accurate with minor omissions*	*Partially correct with missing points*	*Incorrect answers with major gaps*
Clarity	*5%*	*Clear, neat, professional presentation*	*Understandable with minor issues*	*Limited clarity, readability issues*	*Poorly written, difficult to understand*

CO2 AT3

Short Answer

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Date 21/10/26

1. Gamma correction for nighttime Traffic Images.

Technique: Gamma correction

Procedure:

* Read the dark image.
* Apply gamma correction using $S = cr^{\gamma}$ where $\gamma < 1$
* Display the original and enhanced image

Result:

The image becomes brighter, improving the visibility of vehicles and pedestrians.

2 CLAHE for underwater images.

Technique: contrast Limited Adaptive Histogram Equilization (CLAHE).

procedure:

* Read the underwater image
* convert it to LAB color space
* Apply CLAHE to the brightness component.
* convert the image back to RGB.

Result:

Local contrast improves, making coral reefs and underwater structures clearer.

3. median filtering for salt and pepper noise

Technique : median filtering

Procedure :

* Read the noisy image.
* Select 3 x 3 neighbourhood around each pixel
* Replace the center pixel with median neighbouring values.
* Repeat for all pixels

Result :

Salt and pepper noise removed while preserving object boundaries.

4. Laplacian sharpening for PCB image :

Technique : Laplacian sharpening

Procedure :

* Read PCB image and convert into grayscale.
* Apply Laplacian operator detect rapid intensity changes.
* Subtract the Laplacian response for original image
* Display sharpened image

Formula : $g(x,y) = f(x,y) - \nabla^2 f(x,y)$

Result : component edges become sharper and easier identify.

5. Gaussian low pass filtering Astronamical image

Technique: Gaussian low pass filtering

procedure:

* Apply Fourier Transform
* multiply frequency spectrum by filtering.
* Apply the Inverse Fourier Transform to obtain the image.

Result:

* High frequency is reduced improving to visibility of cdestial objects

6. High pass filtering for Textile Images:

Technique: point detection high pass filtering

procedure:

* Apply the fourier Transform
* use high pass filter suppress low frequency background variations.
* Apply the inverse Fourier Transform.

Result:

Fine thread patterns and fabric irregulation become more visible.

7. point and line detection metal surface inspection

Technique: point detection are directional line detection.

Procedure:

* Read the metal surface image and cannot to grayscale
* Apply point detection mask to identify isolated defective pixels.
* mark the detected defects on the image.

Result:

Isolated defects and narrow scratches are metal surface.

8. Adaptive thresholding for Handwritten Answer:

Technique: Adaptive thresholding

Procedure:

* Read answer sheet and convert grayscale
* Divide the image into small local region
* Calculate sepearte threshold for each region

Result:

Handwritten characters are seperated from the background even under uneven lighting.

9) Region - Growing Segmentation for flood Affected Areas.

Technique: Region growing

Procedure:

* Read the satellite image

* Add similar connected pixels to the region.
* continue until no more matching pixels are found.

Result:

connected flood affected regions are extracted from the Satellite Image.

10. Edge Linking and Boundary Detection for Engineering Drawings:

Technique: Edge linking and Boundary detection

Procedure:

* Detect edges using the canny edge detector.
* Apply morphological closing to connect small gaps.
* Detect contours to obtain continuous boundaries.

Result:

Disconnected lines and incomplete object boundaries are reconstructed into continuous structural outlines.